Causal Exaggeration

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Abstract

To avoid confounding, causal identification strategies focus on a subset of the variation in the data, the plausibly exogenous part. In this paper, I argue that this can reduce statistical power and lead published estimates to exaggerate true effects sizes. Using realistic fake data simulations and a mathematical derivation, I show that for the main causal strategies, a perfectly convincing identification does not guaranty an absence of "bias" and that improving identification can actually pull us away from the true effect. I then discuss potential avenues to address this issue.

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I Introduction

One of the main challenges of empirical economics is to reduce confounding to identify causal effects. Identifications strategies such as Regression Discontinuity (RD), Instrumental Variable (IV), Difference-in-Differences (DID) and event studies help us achieve this goal. To do so, these strategies only use part of the variation in the data. They exploit the exogenous part of the variation in the treatment or decrease the sample size by only considering observations for which the as-if random assignment assumption is credible. In this paper, I emphasize that reducing the variation used can decrease precision and thus statistical power, that is to say the probability of rejecting the null hypothesis of no effect when it is false. There is therefore often a tension between reducing confounding and statistical power.

When statistical power is low, not only the estimator is imprecise as we well know but statistically significant estimates exaggerate the true effect size (Ioannidis 2008, Gelman and Carlin 2014, Lu et al. 2019, Zwet and Cator 2021). Only estimates at least two standard errors away from zero are statistically significant at the 5% level. In under-powered studies, these estimates make up a selected sub-sample of all estimates, located in the tails of the distribution of all possible estimates. The average of these statistically significant estimates differs from the true effect, located at the center of the distribution if the estimator is unbiased. Figure 1 illustrates the inflation of significant estimates caused by imprecision. When power is low, obtaining a statistically significant estimate from an unbiased estimator does not guarantee that it will be close to the true effect. An estimate $\hat{\beta}$ of β might be unbiased in the traditional sense of $\mathbb{E}[\hat{\beta}] = \beta$ but conditionally biased in the sense that $\mathbb{E}[\hat{\beta}|\text{ Significant}] \neq \beta$. For statistically significant estimates, the tension between statistical power and reducing confounding is thus a tension between reducing confounding and exaggerating the true effect size.

This consequence of low statistical power could be non-problematic absent two circumstances. First, a large literature underlined the existence of a publication bias favoring statistically significant results in economics (Rosenthal 1979, Andrews and Kasy 2019, Abadie 2020, Brodeur et al. 2020, for instance). Published estimates from under-powered studies thus form a biased sample of the distribution of all estimates of the effect of interest and can greatly exaggerate true effect sizes. Second, the economics literature, as others, suffers from serious power issues. The resulting exaggeration has been documented as participating in the current replication crisis affecting various fields such as economics, epidemiology, medicine or psychology (Button et al. 2013, Open Science Collaboration 2015, Camerer et al. 2016, Chang and Li 2022). Even in experimental economics, with a high level of control and an arguable absence of confounders, estimates published in top economic journals that have been replicated were on average inflated by a factor of at least 1.5 (Camerer et al. 2016). Quasi-experimental studies are likely even more exposed to this exaggera-

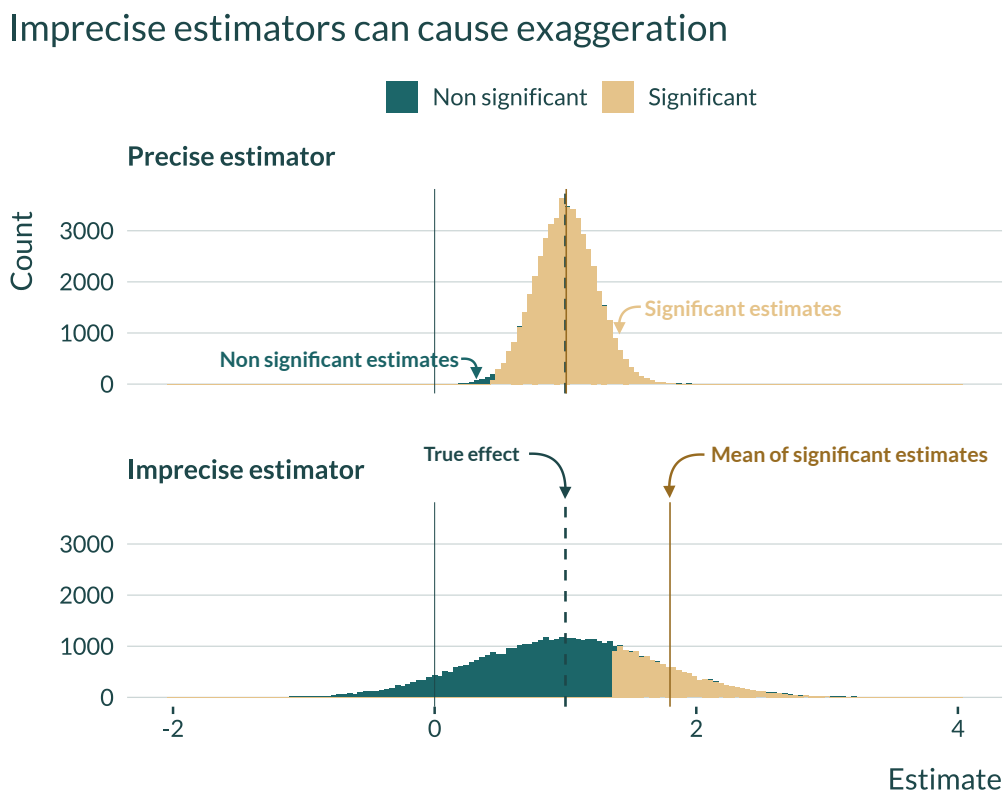


Figure 1 – Significance and distribution of two unbiased estimators with different variances

tion issue as, in current practices, statistical power is not central to such analyses. Ioannidis et al. (2017) concernedly finds that the median statistical power in a wide range of economic studies is no more than 18% and that nearly 80% of estimates may be exaggerated by a factor of two, despite the widespread use of convincing causal identification strategies and usually large sample sizes. The magnitude of exaggeration is thus considerable and in some situations could be on par with that of bias caused by confounders. It is thus crucial to take exaggeration risks into account and to understand its drivers.

In this paper, I propose an overarching mechanism that may contribute to do so. I show that design choices in quasi-experimental studies can be seen as a trade-off between avoiding confounding and overestimating true effect sizes due to a resulting loss in power. To limit the threat of confounding, causal inference methods discard variation and therefore reduce statistical power. This can result in inflated significant and published estimates. While causal identifications strategies are essential to describe causal relationships, this paper emphasizes that a perfectly convincing identification does not guaranty an absence of "bias" and that improving identification can actually pull us away from the true effect. In some settings, reaching unbiasedness can actually be impossible as getting rid of bias caused by confounders creates another type of bias.

While the general idea that causal inference methods discard variation to identify effects is shared across strategies, the confounding / exaggeration trade-off is mediated through a distinctive channel for each of them. In RD designs, even when the initial sample size is large, we discard part of the variation by only considering observations within the bandwidth, decreasing the effective sample size and thus precision. In an IV setting, we only use the subset of the variation in the treatment that is explained by the instrument. In studies leveraging exogenous shocks, the variation used to identify an effect sometimes only comes from a limited number of changes in treatment status. Causal approaches that do not actually leverage natural experiments but aim at identifying a causal effect by controlling for confounders also limit the variation used to identify effects. Matching prunes units that cannot be matched and thus reduces the effective sample size. Generic controls or fixed effects can increase the variance of the estimator and exaggeration if they absorb more of the variation in the treatment than in the outcome variable.

Since causal identification strategies can be interpreted as ways of controlling for confounders, this last point actually ties all the strategy specific arguments together. We implement Fixed Effects (FEs) based identification strategies such as DiD to control for the invariant, unobserved and arguably endogenous part of the variation in the outcome. In the Control Function (CF) approach of IV, we control for the predicted residuals of the regression of the endogenous variable of interest on the instrument. Fuzzy-RD and propensity score matching can also be thought of as control function approaches, of the forcing variable and propensity score respectively. In addition, excluding observations that are outside the bandwidth or unmatched is equivalent to controlling for observation level fixed effects for these observations. When these methods for controlling for confounders absorb more of the variation in the treatment than in the outcome, they will increase the variance and cause exaggeration. Considering a simple linear homoskedastic model gives the intuition for this trade-off between exaggeration and omitted variable bias (OVB) for control approaches. Let $y_i = \alpha + \beta x_i + \delta w_i + u_i$, $\forall i \in \{1, .., n\}$, with w potentially unobserved. Under usual assumptions and using the Frisch-Waugh-Lovell theorem we get that σ^2_{OVB} and σ^2_{CTRL}, the variance of the estimators for β when omitting w (short regression) and controlling for it (long regression) respectively are:

$$\sigma^2_{\text{OVB}} = \frac{\sigma^2_{u_{\text{OVB}}}}{n\ \sigma^2_x} = \frac{\sigma^2_{y^{\perp x}}}{n\ \sigma^2_x} \qquad \text{and} \qquad \sigma^2_{\text{CTRL}} = \frac{\sigma^2_{u_{\text{CTRL}}}}{n\ \sigma^2_{x^{\perp w}}} = \frac{\sigma^2_{y^{\perp x,w}}}{n\ \sigma^2_{x^{\perp w}}}$$

and thus,

$$\sigma^2_{\text{OVB}} < \sigma^2_{\text{CTRL}} \quad \Leftrightarrow \quad \frac{\sigma^2_{y^{\perp x}}}{n\ \sigma^2_x} < \frac{\sigma^2_{y^{\perp x,w}}}{n\ \sigma^2_{x^{\perp w}}} \quad \Leftrightarrow \quad \frac{\sigma^2_{x^{\perp w}}}{\sigma^2_x} < \frac{\sigma^2_{y^{\perp x,w}}}{\sigma^2_{y^{\perp x}}}$$

Controlling will increase the variance of the estimator if the fraction of the variance unexplained

by w is greater for $y^{\perp x}$ than for x. Put differently, if controlling absorbs more of the variation in x than in the residual part of y ($y^{\perp x}$), it will increase the variance of the estimator. As briefly discussed above, since exaggeration increases with the variance of the estimator, controlling may increase exaggeration. I develop a formal proof showing that exaggeration can be larger when controlling, even when accounting for bias, in section III.

In the remainder of the paper, I first illustrate the existence of this "causal exaggeration" using examples drawn from education, labor, environmental and political economics. The exaggeration of statistically significant estimates can be defined as the ratio of the estimated effect over the true effect, which is never known in a real world setting. To know the true effect and be able to compute this quantity, I have to turn to simulations. Since my fake-data Monte-Carlo simulations have an illustrative purpose only, I intentionally focus on settings in which statistical power can be low. All other simulation assumptions are chosen to make it as easy as possible to retrieve the effect of interest. I consider simple linear models with constant treatment effects, i.i.d. observations and homoskedastic errors. All the models are correctly specified and accurately represent the data generating process, except for the omitted variable.

Then, I derive a formal proof of the existence of the trade-off for most prevailing causal identification strategies. Specifically, I show that the bias caused by exaggeration can be larger than the one caused by confounders. I also study the drivers of exaggeration and show that it increases as the strength of the instrument decreases, the number of exogenous shocks decreases or when controlling for an omitted variable absorbs more of the variation in the treatment than in the outcome.

Finally, I discuss concrete avenues to address this causal exaggeration when carrying out a non-experimental study. First, I advocate for the use of tools to evaluate the potential magnitude and risk of both confounding and exaggeration issues separately. Sensitivity analyses help with the former while power calculations help with the latter. For instance, the sensitivity analysis tools developed in Cinelli and Hazlett (2020) enable to assess how strong confounders would have to be to change the estimate of the treatment effect beyond a given level we are interested in. Prospective power simulations help identify the design parameters affecting power and exaggeration by approximating the data generating process (Gelman 2020, Black et al. 2021). Retrospective power calculations allow to evaluate whether a study would have enough power to confidently estimate a range of smaller but credible effect sizes (Gelman and Carlin 2014, Stommes et al. 2021). Focusing more specifically on the trade-off and its drivers, I present tools to visualize the variation actually used for identification when using causal identification strategies. The companion website describes in details how such analyses can be implemented. Finally, I briefly discuss potential solutions to mitigate this trade-off.

This paper contributes to three strands of the literature. First, the idea that causal identifica-

tion estimators, while unbiased, may be imprecise is not new; this is part of the well-known bias / variance trade-off (Imbens and Kalyanaraman 2012, Deaton and Cartwright 2018, Hernán and Robins 2020, Ravallion 2020). I approach this literature from a different angle: through the prism of statistical power and publication bias. Not only the limited precision resulting from the use of causal identification methods could make it difficult to draw clear conclusions regarding the exact magnitude of the effect but I argue that it might also inherently lead to inflated published effect sizes, creating another "bias".

Second, recent studies discussing the exaggeration of statistically significant estimates due to low power focused on specific causal identification methods separately and usually do not investigate the determinants of this exaggeration (Schell et al. 2018, Black et al. 2021, Stommes et al. 2021, Young 2021). In a companion paper, I consider a wide range of causal and non-causal identification strategies and highlight tangible design parameters that can cause exaggeration (Bagilet 2023). In the present paper, I take a step back and propose an overarching mechanism, inherent to causal identification strategies as a whole, and that can explain these issues: although each strategy does so through different means, in essence they discard part of the variation, thereby increasing the risks of exaggeration.

Third, this study contributes to the literature on reproducibility in economics (Camerer et al. 2016, Ioannidis et al. 2017, Christensen and Miguel 2018, Kasy 2021). The trade-off presented in this paper may be an additional explanation for observing replication failures in empirical economics, despite the widespread use of convincing causal identification methods.

II Simulations

First, I define an exaggeration ratio E, as the expected value of significant estimates over the true effect [1]:

Definition 1. *For an estimator $\hat{\beta}$ of a true effect of magnitude β, with standard deviation σ an a two-sided test of size α with threshold value z_α,*

$$E(\hat{\beta}, \sigma, \beta, z_\alpha) = \frac{\mathbb{E}\left[\hat{\beta} \;\middle|\beta, \sigma, |\hat{\beta}| > z_\alpha \sigma\right]}{\beta} \tag{1}$$

This metrics depends on the true effect and can therefore only be computed when this true effect is known. Since it is never the case in real world settings, I have to turn to fake data

1. Note that Gelman and Carlin (2014) define exaggeration as the expected value of the *absolute value* of the estimate: $\mathbb{E}[|\hat{\beta}|| \; \beta, \sigma, |\hat{\beta}| > z_\alpha \sigma]/|\beta|$. Problematically, this expression, when not conditioning on significance, *i.e.* $\mathbb{E}[|\hat{\beta}|| \; \beta, \sigma]/|\beta|$, is not equal to 1 for unbiased estimators, if statistical power is low. The alternative definition I use satisfies this key condition. My exaggeration ratio is also smaller or equal to Gelman and Carlin's and thus more conservative.

Monte-Carlo simulations.

For clarity, I split the simulations by identification strategy and build simulations that reproduce real world examples from economics of education for RDD, labor economics for matching, political economy for IV, health economics for exogenous shocks and environmental economics for control and fixed effects approaches. Real world settings enable to clearly grasp the relationships between the different variables and to set realistic parameter values. Since all these simulations have an illustrative purpose only, I intentionally restrict the simulation exercise to settings in which statistical power can be low. All the models are correctly specified and accurately represent the data generating process, except for the omitted variable bias (OVB).

For each identification strategy, I start by laying out the intuition behind the method and how it enables to estimate causal effects. It naturally points to the key parameter through which the confounding/exaggeration trade-off is mediated. I then briefly describe the example setting considered and the simulation assumptions. I finally display the simulation outputs and discuss the implications of the trade-off that are specific to the identification strategy considered. Detailed codes for simulation procedures are available on the project's website.

II. 1 Regression Discontinuity Design

Intuition for the trade-off. To identify a causal effect, a regression discontinuity approach relies on the assumption that for values close to the threshold, treatment assignment is quasi-random. Under this assumption, individuals just below and just above the threshold would be comparable in terms of observed and unobserved characteristics, and only differ in their treatment status. To avoid confounding, the RDD focuses on observations within a certain bandwidth around the threshold and discards observations further away. The effective sample size where the identification of causal effect of the treatment is the most credible is thus smaller than the total sample size, leading to a lower precision. For this method, the confounding / exaggeration trade-off is therefore mediated by the size of the bandwidth.

Case-study and simulation procedure. To illustrate this trade-off, I consider a standard application of the sharp RD design in economics of education in which students are offered additional lessons based on the score they obtained on a standardized test. Thistlethwaite and Campbell (1960) introduced the concept of RDD using a similar type of quasi-experiment. Students with test scores below a given threshold receive the treatment while those above do not. Since students far above and far below the threshold may differ along unobserved characteristics such as ability, a RDD estimates the effect of the treatment by comparing outcomes of students whose initial test

scores are just below and just above this threshold.

The simulation framework for the RDD is as follows. If a student i has an initial scores $Qual_i$ below a cutoff C, they must take additional lessons, making the allocation of the treatment T sharp: $T_i = \mathbb{I}[Qual_i < C]$. Final scores are correlated with qualification score. Further assume that both qualification and final test scores are affected by students' unobserved ability w in a non-linear (cubic) way. A high or low ability has a strong positive impact on test scores while an average one does not strongly impact test scores. The final test score of student i is thus: $Final_i = \beta_0 + \beta_1 T_i + \eta Qual_i + \delta f(w_i) + u_i$, where f a non linear function (here cubic) and $u_i \sim \mathcal{N}(0, \sigma_u^2)$ random noise. β_1 is the causal parameter of interest.

To make the simulations realistic, I derive parameters values from statistics from the Department of Education and treatment effect sizes from a meta-analysis of RCTs in economics of education by Kraft (2020). Given these parameters values, I then generate 1000 datasets with 10,000 observations. For each dataset, I estimate the treatment effect by regressing the final score on the treatment status and the qualifying score for different bandwidth sizes.

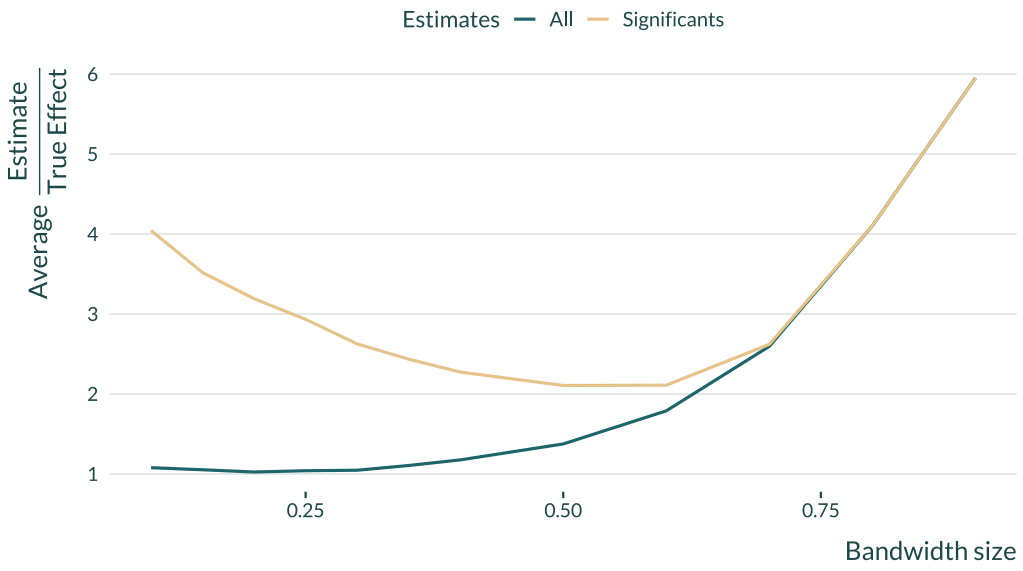
Results. Figure 2 displays the results of these simulations. While the average of all estimates gets close to the true effect as bandwidth size and thus OVB decrease, the average of statistically significant estimates never gets close to the true effect in this setting. For large bandwidths, the omitted variable biases the effect while for small bandwidths, the small sample size creates exaggeration issues. The optimal bandwidth literature describes a similar trade-off but with different consequences (Imbens and Kalyanaraman 2012). They consider a bias/precision trade-off, we consider a omitted variable bias / exaggeration bias trade-off.

II. 2 Instrumental Variable Strategy

Intuition for the trade-off. Instrumental variable strategies overcome the issue of unobserved confounding by only considering the exogenous variation in the treatment, *i.e.* the variation that is explained by the instrument. Even when this exogenous fraction of the variation is limited, the instrument can successfully eliminate confounding on average. However, the IV estimator will be imprecise and statistical power low. In the case of the IV, the confounding / exaggeration trade-off is therefore mediated by the strength of the instrument considered. The weaker the instrument, the more inflated statistically significant estimates will be.

Case-study and simulation procedure. To illustrate this trade-off, I focus on the impact of voter turnout on election results. To avoid the threat of confounding in this setting, studies often take advantage of exogenous factors such as rainfall that affect voter turnout. I reproduce such

Figure 2 – Evolution of the Bias with Bandwidth Size in Regression Discontinuity Design, conditional on significativity.



Notes: The green line indicates the average bias for all estimates, regardless of their statistical significance. The beige line represents the inflation of statistically significant estimates at the 5% level. In this simulation, N = 10,000. The bandwidth size is expressed as the proportion of the total number of observations of the entire sample. Details on the simulation are available at this link.

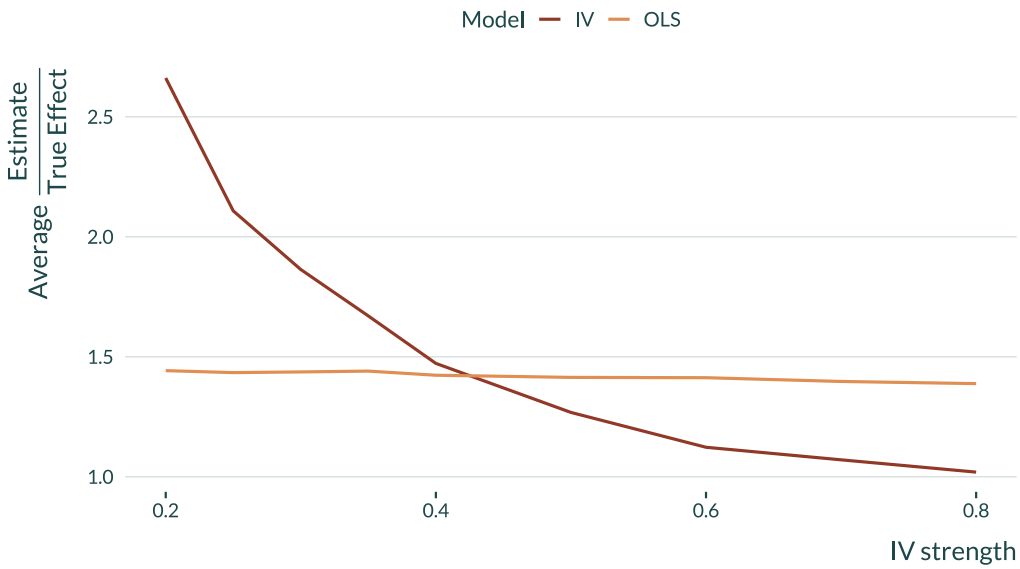
setting and assume that the true data generating process for the republican vote share is such that in location i, $Share_i = \beta_0 + \beta_1 Turnout_i + \delta w_i + u_i$, where w is an unobserved variable and $u \sim \mathcal{N}(0, \sigma_u^2)$ some random noise. The causal parameter of interest is β_1. In addition, turnout is affected by raining events: $Turnout_i = \pi_0 + \pi_1 Rain_i + \gamma w_i + e_i$, where $Rain_i$ is the amount of rain in location i on the day of the election and e some random noise drawn form $\mathcal{N}(0, \sigma_e^2)$. I refer to π_1 as the strength of the instrumental variable.

To make the simulations realistic, I derive parameters values from a set of existing studies using similar variables (Gomez et al. 2007, Fujiwara et al. 2016, Cooperman 2017). For each value of the IV strength considered, I create 1000 datasets. I run both a naive ordinary least squares model and a two-stage least squares model to estimate the impact of voter turnout on the vote share of a party.

Results. Figure 3 displays, for different IV strengths, the average of statistically significant estimates scaled by the true effect size for both the IV and the naive regression model. When the instrument is strong, the IV will recover the true effect, contrarily to the the naive regression model. Yet, when the IV strength decreases, the exaggeration of statistical significant estimates skyrockets. Even if the intensity of the omitted variable bias is large, for limited IV strengths, the exaggeration ratio can become larger than the omitted variable bias. When the only available

instrument is weak, using the naive regression model would, on average, produce statistically significant estimates that are closer to the true effect size than the IV. Of interest for applied research, a large F-statistic does not necessarily attenuate this problem. For the parameter values considered here, this phenomenon arises even in cases for which the F-statistic is substantially larger than the usually recommended threshold of 10, as illustrated in the online supplementary materials.

Figure 3 – Evolution of the Bias of Statistically Significant Estimates Against Strength of the Instrument in the IV Case.



Notes: The red solid line indicates the average bias for IV estimates that are statistically significant at the 5%. The orange line represents the bias of statistically significant OLS estimates at the 5% level. The strength of the instrumental variable is expressed as the value of the linear parameter linking rainfall to turnout. In this simulation, N = 10,000. Details on the simulation are available at this link.

II. 3 Exogenous shocks

Intuition for the trade-off. To avoid confounding, strategies such as DiD and event studies take advantage of exogenous variation in the treatment status caused by exogenous shocks or events. In many settings, while the number of observations may be large, the number of events, their duration or the proportion of individuals affected might be limited. As a consequence, the number of (un)treated observations can be small and the variation available to identify the treatment limited. Statistical power is maximized when the proportion of treated observations is equal to the proportion of untreated ones, as extensively discussed in the randomized controlled trial literature. In studies using discrete exogenous shocks, a confounding / exaggeration trade-off

is thus mediated by the number of observations treated. This issue does not only concern DiD event studies but is particularly salient in this case.

Case-study and simulation procedure. To illustrate this trade-off, I simulate a study of the impact of air pollution reduction on newborn weight of babies. To avoid confounding, one can exploit exogenous shocks to air pollution such as plant closures, creation of a low emission zone or of an urban toll. I simulate the analysis at the zip code and monthly levels and focus on the example of toxic plant closures. I consider that the average birth weight in zip code z at time period t, bw_{zt}, depends on a zip code fixed effect ζ_z, a time fixed effect τ_t, and the treatment status $T_{z,t}$, which is equal to one if a plant is closed in this period and 0 otherwise. The average birth weights bw_{zt} is defined as follows: $bw_{z,t} = \alpha + \beta T_{z,t} + \zeta_z + \tau_t + u_{z,t}$. To further simplify the identification of the effect, I assume a non-staggered treatment allocation and constant and homogenous effects. I only vary the proportion of zip codes affected by toxic plant closings.

The parameters values of the simulations are inspired from Currie et al. (2015) and Lavaine and Neidell (2017). For a fixed sample size of 120,000 observations, I generate 1000 datasets for an increasing number of treated observations and estimate the correct two-way fixed effects model.

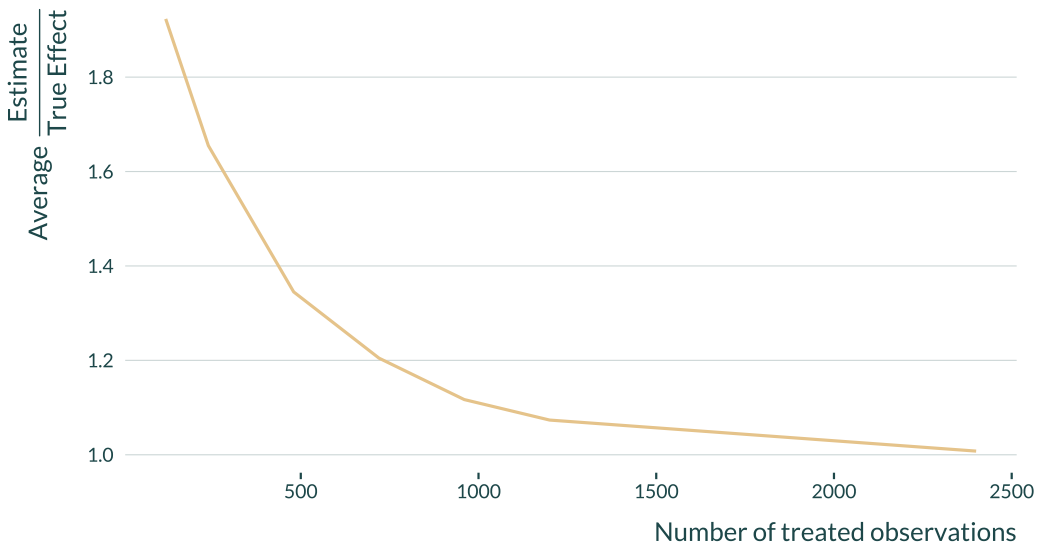
Results. Figure 4 displays the results of these simulations. Even though the actual sample size is extremely large in the example, if the number of treated observations is small, exaggeration can be important. A very large number of observations does not necessarily prevent exaggeration to arise.

II. 4 Matching

Intuition for the trade-off. Then, I focus on the ideal case where all confounders are assumed to be observed. Under this assumption, one can use matching to estimate the causal effect specific to matched treated units. In the case of propensity score matching, observed confounders are adjusted for by predicting the probability of units to take the treatment. Treated units are then matched to control units with similar propensity scores, *i.e.* whose difference is lower than a critical value called the caliper. The smaller the caliper, the more comparable units are and therefore the lower the risk of confounding is. Yet, with a stringent caliper, some units may not be matched, decreasing the sample size. This can lead to a loss in statistical power and produce inflated statistically significant estimates. In the case of matching, the confounding / exaggeration trade-off is therefore mediated by the value of the caliper.

Case-study and simulation procedure. I illustrate this issue by simulating a labor training program where the treatment is not randomly allocated (Dehejia and Wahba 1999). Individuals

Figure 4 – Evolution of Bias With the Number of Treated Observations, for Statistically Significant Estimates, in the Exogenous Shocks Case



Notes: The line indicates the average bias for estimates that are statistically significant at the 5%. In this simulation, N = 120,000. Details on the simulation are available at this link.

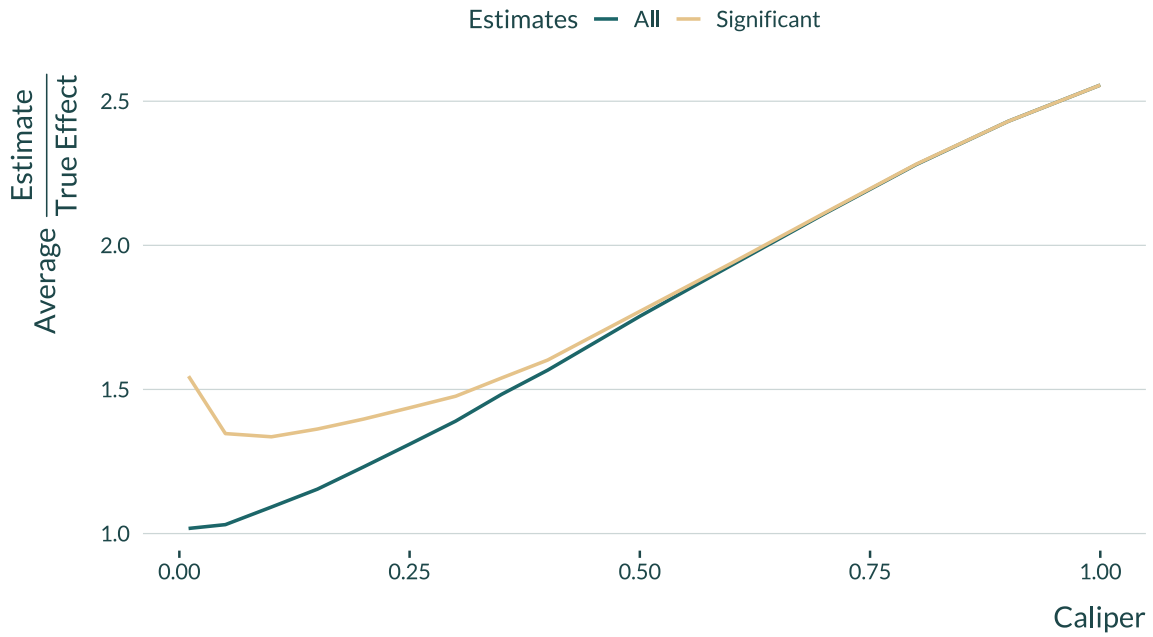
self-select into the training program and may therefore have different characteristics from individuals who do not choose to enroll. To emulate this, I assume that the distribution of the propensity scores differ for treated and control groups: they are drawn from $N(\mu_T, \sigma_T)$ and $N(\mu_C, \sigma_C)$ respectively.

I simulate this type of studies by taking a short-cut. Many simulations found in the applied statistics literature test the performance on matching algorithms by first simulating covariates and then simulating the true but unknown probability of units to be treated. Our goal here is different as we do not want to test the performance of various matching algorithms but rather illustrate how a lack of common overlap in propensity scores can result in a loss of statistical power. We therefore first assign a fraction of individual to the treatment and then simulate the true propensity score variable for treated and control units. For treated units, we draw the propensity scores from a normal distribution $N(\mu_T, \sigma_T)$ and for control units, from $N(\mu_C, \sigma_C)$. Once the true propensity scores are created, we define the potential outcomes of each individual. Here, potential outcomes represent the monthly income (in euros) of the individuals if they undertake the training program or not. The potential outcome of each individual i without treatment adoption, $Y_i(0)$, is simulated using the following equation: $Y_i(0) = \text{Wage} \times \text{PS}_i + N(\mu_\text{N}, \sigma_\text{N})$. Wage is the baseline wage, PS_i the propensity score of individual i and some noise is drawn from $N(\mu_\text{N}, \sigma_\text{N})$. This equation makes the potential outcomes Y(0) partly different for treated and control units, creating the required common support issue. We then simulate the potential outcomes $\text{Y}_i(1)$ by adding a constant

treatment effect of the training program. The constant treatment effect assumption is made to simplify the illustration of the issue we are interested in. In our simulations, when we make the propensity score matching more stringent, not all treated units can be matched to similar control units. The causal estimand would no longer be the average treatment on the treated if the causal effect was not constant across units.

Based on this simulation framework, we generate 1000 datasets for each propensity score matching procedure with caliper values ranging from 0 to 1. Parameters values of the simulation are set to make them realistic and can be found here. Once units are matched, we simply regress the observed revenue on the treatment indicator.

Figure 5 – Evolution of Bias with the Caliper in Propensity Score Matching, Conditional on Statistical Significance.



Notes: The green line indicates the average bias for all estimates, regardless of their statistical significance. The beige line represents the inflation of statistically significant estimates at the 5% level. The caliper is expressed in standard deviation of the propensity score distribution. Details on the simulation are available at this link.

Results. Figure 5 indicates that the average bias of estimates, regardless of their statistical significance, decreases with the value of the caliper as units become more comparable. As the caliper decreases, statistically significant estimates start being more inflated than the entire sample of estimates. For large caliper values, units are not comparable enough and confounding bias the effect. For small caliper values, the sample size becomes too small to be able to precisely estimate the treatment effect and exaggeration arises. It is well-known that matching procedure can result

in imprecise estimates since it does not use information on outcomes but rather focuses on reducing bias arising from covariates imbalance (Rubin 2001). Yet, in a context of publication bias favoring statistically significant estimates, it may make the method produce misleading claims on treatment effect sizes.

III Mathematical derivation

In this section, I formally prove the existence of the confounding / exaggeration trade-off and describe its drivers in a simple setting [2]. To do so, I first show that the exaggeration ratio increases with the variance of normally distributed estimators. This leads me to computing the asymptotic distributions of a series of estimators in order to prove their normality and study drivers of their variances, and ultimately of their exaggeration ratios. Finally, I show that, for any magnitude of OVB, exaggeration can be greater when controlling for confounders using a causal inference methods than the overall bias combining exaggeration and OVB in the naive regression.

III. 1 Properties of the exaggeration ratio

We can easily extend proofs by Lu et al. (2019) and Zwet and Cator (2021) to biased estimators, to show that, for a given size of a test and true effect size, the exaggeration ratio increases with the variance of a (un)biased normally distributed estimator:

Lemma 1. [3]

For an estimator $\hat{\beta}_b \sim \mathcal{N}(\beta + b, \sigma^2)$ of a true effect of magnitude β and a fixed bias b of the same sign as and independent from the true effect,

- *E is a decreasing function of the Signal-to-Noise Ratio (SNR), $\frac{\beta}{\sigma}$, and only depends on σ through this SNR.*
- $\lim_{\sigma \to \infty} E(\hat{\beta}_b, \sigma, \beta, z_\alpha) = +\infty$.

Figure 1 provides a clear intuition for this result in the unbiased case. Note that I focus on cases in which the bias is in the same direction as the true effect to be conservative when comparing exaggeration from causal inference methods to OVB.

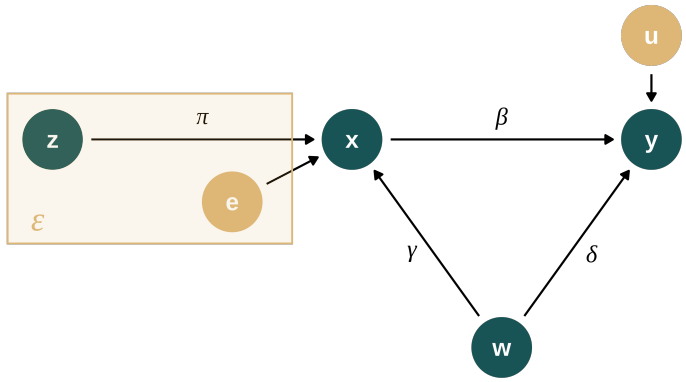
Since the exaggeration ratio only with the variance of the estimator, to study how it evolves with the IV strength in an IV setting, the number of exogenous shocks in a reduced form and the correlation between the explanatory variable of interest and the omitted variable of interest, we only have to study how the variance of these estimators evolves with these parameters.

2. A more detailed version of this mathematical derivation is available on the companion website.
3. All the proofs of the lemma and theorem are in appendix A.

III. 2 Setting and data generating process

Consider an usual linear homoskedastic regression model with an omitted variable. The Directed Acyclic Graph (DAG) in figure 6 represents the data generating process:

Figure 6 – DAG of the data generating process



Notes: for clarity the error terms are represented in this graph, in beige. Model parameters are noted as edges labels.

More specifically, for all individual $i \in \{1, ..., n\}$,

$$y_i = \beta_0 + \beta_1 x_i + \delta w_i + u_i \tag{2}$$

where $y_i \in \mathbb{R}$ is the observed outcome of interest, $x_i \in \mathbb{R}$ the observed explanatory variable of interest, $w_i \in \mathbb{R}$ an unobserved omitted variable, $u_i \in \mathbb{R}$ an unobserved error term. $(\beta_0, \beta_1, \delta) \in \mathbb{R}^3$ are unknown parameters. β_1 is the parameter of interest. For readability, let's introduce the usual vector notation such that for instance $y = (y_1, ..., y_n)'$ and set $\boldsymbol{\beta} = (\beta_0, \beta_1)'$ and $\mathrm{x}_i = (1, x_i)'$.

Assume homogeneous treatment effects and homoskedasticity, along with the usual OLS assumptions (*i.i.d.* observations, finite second moments, positive-definiteness of $\mathbb{E}[\mathrm{x}_i \mathrm{x}_i']$ and u_i conditional mean-zero and uncorrelated with x_i and w_i). Assume that w_i is unobserved, correlated with x_i and that $\delta \neq 0$. To simplify the derivations, I further assume that the unobserved variable is centred, *i.e.* $\mathbb{E}[w_i] = 0$. I also assume that the variance of the component of w_i that is orthogonal to x_i (denoted $w_i^{\perp x}$) does not vary with x_i, *i.e.*, $\mathrm{Var}(w_i^{\perp x}|x_i) = \mathrm{Var}(w_i^{\perp x})$. Consider the following data generating process for x_i:

$$x_i = \mu_x + \gamma w_i + \epsilon_i \tag{3}$$

where $\gamma \in \mathbb{R}^*$ since x and w are correlated. Set $\rho_{xw} = \mathrm{corr}(x, w) = \frac{\gamma \sigma_w}{\sigma_x}$. In the IV and reduced form sections, I further assume that there exists a valid instrumental variable z_i for x_i, *i.e.* that $\mu_x + \epsilon_i = \pi_0 + \pi_1 z_i + e_i$ where $(\pi_0, \pi_1) \in \mathbb{R}^2$ are unknown parameters. The existence or not of this

valid instrument does not affect the results in the controled and OVB cases. Since the instrument is valid, it satisfies exogeneity, *ie* $\mathbb{E}[\mathrm{z}_i u_i] = 0$ and $\mathbb{E}[z_i w_i] = 0$, relevance, *ie* $\mathrm{rank}(\mathbb{E}[\mathrm{z}_i \mathrm{x}_i']) = 2$, and positive-definiteness of $\mathbb{E}[\mathrm{z}_i \mathrm{z}_i']$. The data generating process for x_i becomes:

$$x_i = \pi_0 + \pi_1 z_i + \gamma w_i + e_i \tag{4}$$

I assume that e_i is uncorrelated with z_i and w_i, *ie* $\mathbb{E}[z_i e_i] = 0$ and $\mathbb{E}[w_i e_i] = 0$. I also assume homoskedasticity for this term, such that $\mathbb{E}[e_i^2 | z_i, w_i] = \sigma_e^2$ is constant.

Overall, this DGP is close to the usual textbook one but with an additional omitted variable.

III. 3 Asymptotic distributions of the estimators

I now derive the asymptotic distributions of the various estimators. The goal is to show asymptotic normality and to study the evolution of estimator variances with the value of the parameter of interest for each model, *i.e.*, a measure of the correlation between x and w (γ) in the controled case, of the IV strength (π_1) in the IV case and of the number of exogenous shocks (σ_z^2 when z is a dummy) in the reduced form case. In order for the variation of one factor not to impact other factors of interest, I consider the variances of the variables ($\sigma_y^2, \sigma_x^2, \sigma_w^2$ and σ_z^2) as fixed but adjust for the variances of the error terms (σ_u^2 and σ_ϵ^2) when varying the values of one of the parameters (γ, δ and π_1). This corresponds to thinking in terms of shares of the variance of x and y explained by "defined" variables (*i.e.*, observed variables and w) *versus* by residuals. Finally, comparison between cases with and without OVB for different parameter values is only relevant if varying the parameter of interest does not affect the OVB. I thus make comparative statics analyses at bias fixed, *i.e.*, as shown below, for $\gamma\delta = \kappa = cst$.

III. 3.1 Naive regression (OVB)

First, consider a "naive" regression of y on x (with w omitted).

Lemma 2. *Based on the data generating process described in section III. 2, for* $\hat{\beta}_{\text{OVB}}$ *the OLS estimate of* β_1 *in the regression of* y *on* x, $\hat{\beta}_{\text{OVB}} \xrightarrow{d} \mathcal{N}(\beta_1 + b_{\text{OVB}},\ \sigma^2_{\text{OVB}})$, *with*

$$b_{\text{OVB}} = \frac{\delta\gamma\sigma_w^2}{\sigma_x^2} \qquad \textit{and} \qquad \sigma^2_{\text{OVB}} = \frac{\sigma_u^2 + \delta^2\sigma_w^2(1-\rho_{xw}^2)}{n\ \sigma_x^2}$$

The intuition for the formula of the asymptotic variance has been discussed in the introduction: $\sigma_u^2 + \delta^2\sigma_w(1-\rho_{xw}^2)$ is the part of the variance in y that is not explained by x ($\sigma^2_{y^{\perp x}}$).

Note that, varying the parameter of interest, ρ_{xw}, will change the bias and σ_u^2. Since σ_x^2 and σ_w^2 are fixed, reasoning at $b_{\text{OVB}} = cst$ is equivalent to considering that $\gamma\delta = \kappa = cst$. Then, noting that $\forall i, u_i = y_i - \beta_0 - \beta_1 x_i - \delta w_i$ and computing its variance, we can rewrite the variance of the estimator as a function of fixed variances and one or less varying parameter:

$$\sigma^2_{\text{OVB}} = \frac{\sigma_y^2 - \beta_1^2\sigma_x^2 - 2\beta_1\kappa\sigma_w^2 - \kappa^2\frac{\sigma_w^4}{\sigma_x^2}}{n\ \sigma_x^2}$$

This expression underlines that, for a given bias, σ^2_{OVB} does not vary with γ, or equivalently δ, the parameters of interest. Applying lemma 1 proves that E_{OVB} does not either.

III. 3.2 Controlled regression

Next, let's then turn to the "ideal" case in which no variable is omitted, *i.e.* we control for the omitted variable w. The model considered accurately represents the DGP. This corresponds to the usual OLS setting with a constant and two regressors that are uncorrelated with the error: y regressed on x and w.

Lemma 3. *Based on the sus-mentioned data generating process, for $\hat{\beta}_{\text{CTRL}}$ the OLS estimator of β_1 in the regression of y on x and w, $\hat{\beta}_{\text{CTRL}} \xrightarrow{d} \mathcal{N}(\beta_1,\ \sigma^2_{\text{CTRL}})$, with*

$$\sigma^2_{\text{CTRL}} = \frac{\sigma_u^2}{n\ \sigma_x^2(1-\rho_{xw}^2)}$$

Note that $\sigma_x^2(1-\rho_{xw}^2)$ is the part of the variance of x that is not explained by w ($\sigma^2_{x^{\perp w}}$) and σ_u^2 the part of the variance of y that is not explained by x nor w ($\sigma^2_{y^{\perp x,w}}$); here too we retrieved a result described in introduction. For a given bias, we then rewrite σ^2_{CTRL} as a function of fixed variances and one varying parameter, γ:

$$\sigma^2_{\text{CTRL}} = \frac{\sigma_y^2 - \beta_1^2\sigma_x^2 - \frac{\kappa^2}{\gamma^2}\sigma_w^2 - 2\beta_1\kappa\sigma_w^2}{n\ (\sigma_x^2 - \gamma^2\sigma_w^2)}$$

Since the numerator and denominator respectively increases and decrease with $|\gamma|$, σ^2_{CTRL} increases with $|\gamma|$. For a given bias, the more w is correlated with x (and thus roughthly the less it is with y since $\delta\gamma = cst$), the larger the variance of the estimator. In addition, we can note that, for a given bias, the variance of the estimator can be arbitrarily large since $\lim_{\gamma^2 \to \frac{\sigma_x^2}{\sigma_w^2}} \sigma^2_{\text{CTRL}} = +\infty$.

III. 3.3 Instrumental Variable

We now want to estimate the IV model in which we regress y on x instrumented by z. We instrument $\mathrm{x}_i = (1, x_i)'$ with $\mathrm{z}_i = (1, z_i)'$. We are thus in a just-identified case and $\hat{\boldsymbol{\beta}}_{\text{2SLS}} = \hat{\boldsymbol{\beta}}_{\text{IV}}$.

Lemma 4. *Based on the sus-mentioned data generating process, for $\hat{\beta}_{\text{IV}}$ the IV estimator of β_1 in the regression of y on x instrumented by z, $\hat{\beta}_{\text{IV}} \xrightarrow{d} \mathcal{N}(\beta_1,\ \sigma^2_{\text{IV}})$, with*

$$\sigma^2_{\text{IV}} = \frac{\sigma^2_u + \delta^2\sigma^2_w}{n\ \sigma^2_x\rho^2_{xz}}$$

Note that the numerator is $\sigma^2_{y\perp\hat{x}}$, the part of the variance in y that is not explained by $\hat{x}$, the predicted value of x in the first stage and the denominator is $\sigma^2_{\hat{x}}$. For a given bias, noting that $\rho_{xz} = \text{corr}(x, z) = \pi_1\frac{\sigma_z}{\sigma_x}$ and replacing σ^2_u, we can rewrite σ^2_{IV} as a function of fixed variances and one varying parameter, π_1:

$$\sigma^2_{\text{IV}} = \frac{\sigma^2_y - \beta^2_1\sigma^2_x - 2\beta_1\kappa\sigma^2_w}{n\ \pi^2_1\sigma^2_z}$$

Clearly, the smaller π_1, the larger σ^2_{IV}. In addition, $\lim_{\pi_1\to 0}\sigma^2_{\text{IV}} = +\infty$.

III. 3.4 Reduced form

Let's now assume that we want to directly estimate the effect of the instrument on the outcome of interest. Plugging equation 4 into equation 2 yields:

$$y_i = (\beta_0 + \beta_1\pi_0) + (\beta_1\pi_1)z_i + ((\delta + \beta_1\gamma)w_i + u_i + \beta_1 e_i)$$

Note that if we directly regress the outcome on the instrument, the resulting estimand will be different from that of the other models. To make them comparable, we could set π_1 to 1 so that an increase of 1 in the instrument causes an increase of β_1 in y. Regardless of whether we make this assumption or not, regressing y on z corresponds to the usual univariate, unbiased case and directly gives the following result:

Lemma 5. *Based on the sus-mentioned data generating process, for $\hat{\beta}_{\text{RED}}$, the OLS estimator of the reduced form regression of y on z, $\hat{\beta}_{\text{RED}} \xrightarrow{d} \mathcal{N}(\beta_1,\ \sigma^2_{\text{RED}})$, with*

$$\sigma^2_{\text{RED}} = \frac{\sigma^2_y - \beta^2_1\pi^2_1\sigma^2_z}{n\ \sigma^2_z}$$

Note that the numerator is the part of the variance of y that is not explained by z ($\sigma^2_{y\perp z}$). In addition, it is clear that the smaller σ^2_z, the larger σ^2_{RED}. In addition, $\lim_{\sigma_z\to 0}\sigma^2_{\text{RED}} = +\infty$.

In the binary case, $\sigma_z^2 = p_1(1-p_1)$ with p_1 the proportion of treated observations, *i.e.*, the proportion of 1 in z. When most observations have the same treatment status, *i.e.*, p_1 close to 0 or 1, σ_z^2 tends to zero and σ_{RED}^2 shoots up. There is not enough variation in the treatment status to precisely identify the effect of interest.

III. 4 Exaggeration ratios

Combining the results from lemma 2 through 5 regarding the asymptotic distribution of the various estimators with lemma 1 stating that exaggeration increases with the variance of a normally distributed estimator yields:

Theorem 1. *For the data generating process described in section III. 2, the exaggeration ratio of the controled, IV and reduced form estimators, respectively $E_{\text{CTRL}}, E_{\text{IV}}$ and E_{RED}, are such that:*

- *E_{CTRL} increases with the correlation between the omitted variable and the explanatory variable of interest (i.e. $|\gamma|$ or $|\rho_{xw}|$), for a given bias,*
- *E_{IV} increases with the strength of the IV (i.e. with $|\pi_1|$ or $|\rho_{xz}|$),*
- *E_{RED} increases when the number of exogenous shocks decreases in the binary case*

Also using the same lemma and the limit properties of the variances described in section III. 2, and since, at fixed bias, E_{OVB} does not vary with the parameters of interest, we get:

Theorem 2. *For the data generating process described in section III. 2,*

- $\forall\ b_{\text{OVB}}, \exists \gamma$ *s.t.* $E_{\text{CTRL}} > E_{\text{OVB}}$
- $\exists\ \pi_1$ *s.t.* $E_{\text{IV}} > E_{\text{OVB}}$
- $\exists\ \sigma_z$ *s.t.* $E_{\text{RED}} > E_{\text{OVB}}$

For some parameter values, statistically significant estimates can be larger on average when using a convincing causal identification strategy that eliminates the omitted variable bias than when embracing the bias and running a naive biased regression.

IV Practical Recommendations

We have seen that using causal methods induces a trade-off between avoiding confoundings and exaggerating true effects if the study is under-powered. How can we, as applied researchers using observational data, arbitrate it? Since key pieces of information such as the true effect and the effect of omitted variables are inherently unknown, we cannot directly compute the biases caused by confounders and exaggeration. In this section, I examine how we can however get a sense of threats from both sides of the trade-off and visualize its main driver, the variation used for identification.

I then discuss how changing attitudes towards statistical significance and replicating studies could limit the exaggeration issue.

IV. 1 Gauging omitted variable bias

On one side of the trade-off lies the bias caused by confounders. Although it is in essence impossible to measure, tools such as sensitivity analyses are available to gauge its magnitude (Rosenbaum 2002, Middleton et al. 2016, Oster 2019, Cinelli and Hazlett 2020). For instance, the method developed in Cinelli and Hazlett (2020) enables to assess how strong confounders would have to be to change the estimate of the treatment effect beyond a given level we are interested in. It offers bounds for the strength of the association between the treatment and potential omitted variables by weighting it against the measured association between the treatment and observed covariates. A typical conclusion from such an analysis would be: "omitted variables would have to explain as much residual variance of the outcome and the treatment as the observed covariate x to bring down the estimate to a value of β_l". In addition, the authors implement graphical tools to facilitate this comparison. I suggest to use such quantitative bias analyses to evaluate the restrictiveness of the causal approach required to limit the threat of unobserved confounding to acceptable levels.

IV. 2 Evaluating risks of exaggeration

On the other side of the trade-off lies the exaggeration emerging when statistical power is low. As OVB, exaggeration and statistical power are in essence impossible to measure as their computation depends on the true effect which is always unknown. Yet, power calculations can help assess them by making hypothesizes on the magnitude of the true effect. In randomized controlled trials, such computations are not only an established practice but a requirement (Duflo et al. 2007, McConnell and Vera-Hernandez 2015, Athey and Imbens 2016). They are, however, rarely reported in non-experimental studies. This can be understandable if power calculations are perceived only as a way to measure a design's ability to detect an effect when there is actually one. Yet, taking publication bias and the threat of exaggeration into account highlights the necessity of running power calculations in non-experimental studies as well. A low power or a relatively large variance not only makes it more difficult to detect an effect or to draw clear conclusions about its magnitude when detected but it can also create a bias. Considering the attention given to bias avoidance in the economics literature, I advocate to make power central to non-experimental analyses to limit bias caused by exaggeration. Currently, in causal inference textbooks, very few pages are devoted to statistical power in non-experimental studies (Angrist and Pischke 2009; 2014, Imbens and Rubin 2015, Cunningham 2021). To the best of my knowledge, only two textbooks

discuss the matter in depth (Shadish et al. 2002, Huntington-Klein 2021). Results from power and exaggeration calculations would not only be highly informative but could also be reported very concisely in the robustness section of articles.

IV. 2.1 Prospective power calculations

To evaluate the statistical power of a study, the risk of exaggeration and identifying the factors driving it, one can first simulate the design of the study (Hill 2011, Gelman 2020, Black et al. 2021). Simulating a data generating process from scratch requires thinking about the distribution of the variables, about their relationships and can also help underline the variation used for identification. I implemented such fake-data simulations in Section II and provide `R` code to be used as examples in the replication material. In situations where the relationships among covariates are too complex to emulate, one can also start from an existing dataset and add a known treatment effect. I implemented real data simulations in a companion paper and describe their implementation in its replication material (Bagilet 2023).

IV. 2.2 Retrospective power calculations

Running post-analysis power calculations can also help getting a sense of the statistical power associated with a research design. Such *retrospective* calculations allow to evaluate whether the design of the study would produce accurate and uninflated statistically significant estimates if the true effect was in fact smaller than the observed estimate (Gelman and Carlin 2014, Ioannidis et al. 2017, Stommes et al. 2021).

I illustrate how a retrospective analysis works by taking the example of Card (1993) on the relationship between human capital and income. He finds that an additional year of education, instrumented by the distance of growing near a four-year college, causes a 13.2% average increase in wage. The associated standard error is 5.5%. Is there a risk of exaggeration with this design? Since, as noted by the author himself, the estimate is very imprecise we could expect so. Imagine that prior evidence suggests that the true effect is likely close to a 10% increase in wage. We can then compute the statistical power of the study by drawing many estimates from a normal distribution centered around the hypothesized true effect of 10% and with a standard deviation equal to the 5.5% standard error obtained in Card (1993). Concretely, this proceeds as if we were able to replicate the study many times under the assumption that the true effect was equal to the hypothesized one [4]. Statistically significant estimates (at the 5% level) would on average be roughly equal to 15%, therefore overestimating the true effect by a factor of 1.5. Statistical

4. Timm et al. (2019) and Linden (2019) offer `R` and `Stata` packages that enable to easily run these calculations through a short command: `retrodesign(10, 5.5)`.

power, the proportion of estimates that are significant, would be only 44%. Conditional on a 10% true effect size being a reasonable assumption, this study would be under-powered and the risk of exaggeration large.

The usefulness of any retrospective power analysis lies on the assumption regarding the true effect size. Informed guesses about the range of plausible effect sizes can be based on results from meta-analyses or previous studies with a convincing design (*e.g.*, a large randomized controlled trial)[5]. When such information is not available, power calculations can be ran for a range of smaller but credible effect sizes.

IV. 3 Driver of the trade-off

We have seen that the variation used for identification was the main driver of the confounding / exaggeration trade-off. This variation determines the variance of the estimator and thus the amount of exaggeration. In this section, I first discuss the impact of estimator variance and how exaggeration leads to reinterpret the well-known bias/variance trade-off. I then propose to use a tool that enables to visualize where the variation used for identification comes from.

IV. 3.1 Navigating the bias/variance trade-off

In non-experimental studies, estimator variance is often important to the extent that a large variance may lead to a failure to reject the null of zero effect when it is actually incorrect. It makes variance paramount as long as a statistically significant estimate is not obtained. Yet, exaggeration underlines that variance matters, even after that a significant estimate has been obtained.

Obtaining a statistically significant estimate from an imprecise estimator should not necessarily be interpreted as a sign of "success" in getting significance despite the estimator being imprecise. It could however be a warning that this estimate comes from the tails of the distribution and thus inaccurately represents the true effect. Conditional on having obtained a statistically significant estimate, a limited precision can hide a bias: exaggeration. This invites to revisit the well-known bias/variance trade-off: a larger variance can actually lead to a larger (conditional) bias. The bias/variance trade-off can in fact be a bias/bias trade-off. This paper thus invites to pay more attention to the implications of our design choices on the variance of our estimators, even after we obtained statistically significant estimates.

5. Note that when such meta-analyses are available, one can use a Bayesian procedure to shrink statistically significant estimates based on the corpus of estimates from prior studies Zwet and Gelman (2021), Zwet and Cator (2021), Zwet et al. (2021) .

IV. 3.2 Visualizing the driver of the trade-off

Causal inference strategies only leverage a subset of the variation to avoid confounders. However, when this subset is too small, exaggeration arises. Visualizing this variation (and the observations) actually used for identification can help navigate the confounding / exaggeration trade-off. The visualization tool I outline here both leverages the interpretation of causal inference methods as control approaches and builds on a procedure developed in Aronow and Samii (2016) for a different purpose: evaluating external validity of standard regressions.

Aronow and Samii (2016) essentially interprets the estimate of the coefficient of the treatment of interest in a simple linear non-causal regression as a weighted average of individual treatment effects. The weight w_i of individual i is simply the squared difference between its treatment status T_i and the value of this treatment status as predicted by the other covariates X: $w_i = (T_i - \mathbb{E}[T_i|X_i])^2$. If treatment effects are heterogenous, the weighting may lead some observations to be disproportionally (non-)represented in the average effect of the treatment. In that case, the average of treatment is only representative of a subset of the individual treatments, leading to external validity issues.

The parallel with our setting directly follows from this interpretation regardless of whether the treatment is heterogenous or not: observations whose treatment status is well explained by covariates do not actually contribute to the estimation of the effect of the treatment. This may lead to a small *effective* sample size and to exaggeration. In the control approach to causal inference strategies, the more variation in the treatment is absorbed when "controlling" for confoundings, the smaller the effective sample, potentially leading to exaggeration. Aronow and Samii (2016) focused on the representativity of the effective sample for fear of external validity issues. I am interested on in its size for fear of exaggeration.

It might then seem compelling to define this effective sample by proposing a weight value under which the associated observation does not actually contribute to identification. Yet, considering the specificity of each analysis and that exaggeration depends on several factors, including the true effect size, I avoid proposing an *ad hoc* cut-off. Instead, I suggest to visualize the individual weights. It allows to get a sense of where the variation comes from and which are the observations that actually contribute to the estimation. Since applied economics analyses often rely on panel data, I propose to use a heatmap as a base for visualization, with time on the x-axis and individuals on the y-axis. Figure displays this visualization for the paper

Most of the variation is absorbed by fixed effects, leading only a small subset of observation to actually contribute to the estimation (the darker ones).

If the data is cross sectional, the same heatmap can be computed with individuals on the x-axis and a length-one segment on the y-axis. I display such graphs and share `R` code to build on the

companion website.

IV. 4 Attitude Towards Statistical Significance and Replication

Exaggeration only arises in the presence of publication bias. As shown in the simulations, if estimates were not filtered by their statistical significance, even under-powered studies would on average recover the true effect, as long as the estimator is unbiased. The exaggeration issue could there be addressed by tackling publication bias.

To identify broader pathways to eliminate this filtering of significant results, it is first helpful to discuss the processes that lead to statistically significant results when power and thus the probability of obtaining a significant estimate is low. In such situations, they can be obtained either by "chance" or as an outcome of the garden of forking paths (Simmons et al. 2011, Gelman and Loken 2013, Kasy 2021). Forks appear at various stages along the path of research, for instance in data preparation, regarding the inclusion of a given control variable or whether to carry on a research that yields non-significant results. Due to the structural flaw that favors significance, the path followed may be more likely to lead to a statistically significant result. These choices are most often not the result of bad researcher practices but instead a product of a structure that portrays significant results as the end goal of research.

The issue being structural, system level changes in scientific practices could also alleviate exaggeration and the trade-off described in this paper. First, many researchers advocate abandoning statistical significance as a measure of a study's quality (McShane et al. 2019). To be effective, this change should be paired with an effort to replicate studies (Christensen and Miguel 2018). Replications, even of low powered studies, would eventually enable to build the actual distribution of the causal estimand of interest. Meta-analyzes would then reduce the uncertainty around the true value of the causal estimand by pooling estimates (Hernán 2021). Finally, the inflation of statistically significant estimates could be limited by considering confidence intervals as compatibility intervals (Shadish et al. 2002, Amrhein et al. 2019, Romer 2020). The width of these intervals gives a range of effect sizes compatible with the data. Confidence intervals will be wide in under-powered studies signaling that point estimates should not be taken at face value, even if statistically significant.

V Conclusion

The economic literature suffers from an extensive lack of statistical power (Ioannidis et al. 2017) and many studies showed that this literature strongly favors statistically significant findings

(Rosenthal 1979, Andrews and Kasy 2019, Abadie 2020, Brodeur et al. 2020, for instance). It has been shown that in such situations, estimates published from underpowered studies exaggerate true effect sizes, even when the estimators are "unbiased" in the usual sense of $\mathbb{E}[\hat{\beta}] = \beta$ (Ioannidis 2008, Gelman and Carlin 2014, Lu et al. 2019, Zwet and Cator 2021). It is therefore not surprising that many estimates published in economics have been shown to be considerably exaggerated (Camerer et al. 2016, Ioannidis et al. 2017), despite the extensive use of convincing causal inference methods. However, determinants for these exaggeration and power issues have remained understudied. I argue that exaggeration may be exacerbated by the foundational component of causal inference: the fact that it only leverages subsets of the variation. Although causal methods enable to avoid confounding, they also reduce statistical power and thus increase the risk of exaggeration. The same aspect that makes these methods credible can create another type of bias. A systematic reporting of statistical power calculations and analysis of the variation actually used for identification could help avoid falling into this exaggeration trap.

References

Abadie, A. (2020), 'Statistical Nonsignificance in Empirical Economics', American Economic Review: Insights **2**(2), 193–208. 2, 25

Amrhein, V., Trafimow, D. and Greenland, S. (2019), 'Inferential Statistics as Descriptive Statistics: There Is No Replication Crisis if We Don't Expect Replication', The American Statistician **73**(sup1), 262–270. 24

Andrews, I. and Kasy, M. (2019), 'Identification of and Correction for Publication Bias', American Economic Review **109**(8), 2766–2794. 2, 25

Angrist, J. D. and Pischke, J.-S. (2009), Mostly Harmless Econometrics: An Empiricist's Companion, 1 edition edn, Princeton University Press, Princeton. 20

Angrist, J. D. and Pischke, J.-S. (2014), Mastering 'Metrics: The Path from Cause to Effect, Princeton University Press. 20

Aronow, P. M. and Samii, C. (2016), 'Does Regression Produce Representative Estimates of Causal Effects?', American Journal of Political Science **60**(1), 250–267. 23

Athey, S. and Imbens, G. (2016), 'The Econometrics of Randomized Experiments', arXiv:1607.00698 [econ, stat] . 20

Bagilet, V. (2023), 'Accurately Estimating Relatively Small Effects: Air Pollution and Health'. 6, 21

Black, B. S., Hollingsworth, A., Nunes, L. and Simon, K. I. (2021), Simulated Power Analyses for Observational Studies: An Application to the Affordable Care Act Medicaid Expansion, SSRN Scholarly Paper ID 3368187, Social Science Research Network, Rochester, NY. 5, 6, 21

Brodeur, A., Cook, N. and Heyes, A. (2020), 'Methods Matter: P-Hacking and Publication Bias in Causal Analysis in Economics', American Economic Review **110**(11), 3634–3660. 2, 25

Button, K. S., Ioannidis, J. P. A., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S. J. and Munafò, M. R. (2013), 'Power failure: Why small sample size undermines the reliability of neuroscience', Nature Reviews Neuroscience **14**(5), 365–376. 2

Camerer, C. F., Dreber, A., Forsell, E., Ho, T.-H., Huber, J., Johannesson, M., Kirchler, M., Almenberg, J., Altmejd, A., Chan, T., Heikensten, E., Holzmeister, F., Imai, T., Isaksson, S., Nave, G., Pfeiffer, T., Razen, M. and Wu, H. (2016), 'Evaluating replicability of laboratory experiments in economics', Science **351**(6280), 1433–1436. 2, 6, 25

Card, D. (1993), Using Geographic Variation in College Proximity to Estimate the Return to Schooling, Working Paper 4483, National Bureau of Economic Research. 21

Chang, A. C. and Li, P. (2022), 'Is Economics Research Replicable? Sixty Published Papers From Thirteen Journals Say "Often Not"', Critical Finance Review **11**. 2

Christensen, G. and Miguel, E. (2018), 'Transparency, Reproducibility, and the Credibility of Economics Research', Journal of Economic Literature **56**(3), 920–980. 6, 24

Cinelli, C. and Hazlett, C. (2020), 'Making sense of sensitivity: Extending omitted variable bias', Journal of the Royal Statistical Society: Series B (Statistical Methodology) **82**(1), 39–67. 5, 20

Cooperman, A. D. (2017), 'Randomization Inference with Rainfall Data: Using Historical Weather Patterns for Variance Estimation', Political Analysis **25**(3), 277–288. 9

Cunningham, S. (2021), Causal Inference: The Mixtape, Yale University Press. 20

Currie, J., Davis, L., Greenstone, M. and Walker, R. (2015), 'Environmental Health Risks and Housing Values: Evidence from 1,600 Toxic Plant Openings and Closings', American Economic Review **105**(2), 678–709. 11

Deaton, A. and Cartwright, N. (2018), 'Understanding and misunderstanding randomized controlled trials', Social Science & Medicine **210**, 2–21. 6

Dehejia, R. H. and Wahba, S. (1999), 'Causal Effects in Nonexperimental Studies: Reevaluating the Evaluation of Training Programs', Journal of the American Statistical Association **94**(448), 1053–1062. 11

Duflo, E., Glennerster, R. and Kremer, M. (2007), Using Randomization in Development Economics Research: A Toolkit, in T. P. Schultz and J. A. Strauss, eds, 'Handbook of Development Economics', Vol. 4, Elsevier, pp. 3895–3962. 20

Fujiwara, T., Meng, K. and Vogl, T. (2016), 'Habit Formation in Voting: Evidence from Rainy Elections', American Economic Journal: Applied Economics **8**(4), 160–188. 9

Gelman, A. (2020), Regression and Other Stories, Cambridge University Press, Cambridge New York, NY Port Melbourne, VIC New Delhi Singapore. 5, 21

Gelman, A. and Carlin, J. (2014), 'Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors', Perspectives on Psychological Science **9**(6), 641–651. 2, 5, 6, 21, 25

Gelman, A. and Loken, E. (2013), 'The garden of forking paths: Why multiple comparisons can be a problem, even when there is no "fishing expedition" or "p-hacking" and the research hypothesis was posited ahead of time'. 24

Gomez, B. T., Hansford, T. G. and Krause, G. A. (2007), 'The Republicans Should Pray for Rain: Weather, Turnout, and Voting in U.S. Presidential Elections', The Journal of Politics **69**(3), 649–663. 9

Hernán, M. A. (2021), 'Causal analyses of existing databases: No power calculations required', Journal of Clinical Epidemiology p. S0895435621002730. 24

Hernán, M. A. and Robins, J. M. (2020), Causal Inference: What If, boca raton: chapman & hall/crc edn. 6

Hill, J. L. (2011), 'Bayesian Nonparametric Modeling for Causal Inference', Journal of Computational and Graphical Statistics **20**(1), 217–240. 21

Huntington-Klein, N. (2021), The Effect: An Introduction to Research Design and Causality, 1 edn, Chapman and Hall/CRC, Boca Raton. 21

Imbens, G. and Kalyanaraman, K. (2012), 'Optimal Bandwidth Choice for the Regression Discontinuity Estimator', The Review of Economic Studies **79**(3), 933–959. 6, 8

Imbens, G. W. and Rubin, D. B. (2015), Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction, Cambridge University Press, Cambridge. 20

Ioannidis, J. P. A. (2008), 'Why Most Discovered True Associations Are Inflated', Epidemiology **19**(5), 640–648. 2, 25

Ioannidis, J. P. A., Stanley, T. D. and Doucouliagos, H. (2017), 'The Power of Bias in Economics Research', The Economic Journal **127**(605), F236–F265. 3, 6, 21, 24, 25

Kasy, M. (2021), 'Of Forking Paths and Tied Hands: Selective Publication of Findings, and What Economists Should Do about It', Journal of Economic Perspectives **35**(3), 175–192. 6, 24

Kraft, M. A. (2020), 'Interpreting Effect Sizes of Education Interventions', Educational Researcher **49**(4), 241–253. 8

Lavaine, E. and Neidell, M. (2017), 'Energy Production and Health Externalities: Evidence from Oil Refinery Strikes in France', Journal of the Association of Environmental and Resource Economists **4**(2), 447–477. 11

Linden, A. (2019), 'RETRODESIGN: Stata module to compute type-S (Sign) and type-M (Magnitude) errors', Boston College Department of Economics. 21

Lu, J., Qiu, Y. and Deng, A. (2019), 'A note on Type S/M errors in hypothesis testing', British Journal of Mathematical and Statistical Psychology **72**(1), 1–17. 2, 14, 25, 31

McConnell, B. and Vera-Hernandez, M. (2015), Going beyond simple sample size calculations: A practitioner's guide, Technical report, Institute for Fiscal Studies. 20

McShane, B. B., Gal, D., Gelman, A., Robert, C. and Tackett, J. L. (2019), 'Abandon Statistical Significance', The American Statistician **73**(sup1), 235–245. 24

Middleton, J. A., Scott, M. A., Diakow, R. and Hill, J. L. (2016), 'Bias Amplification and Bias Unmasking', Political Analysis **24**(3), 307–323. 20

Open Science Collaboration (2015), 'Estimating the reproducibility of psychological science', Science **349**(6251), aac4716. 2

Oster, E. (2019), 'Unobservable Selection and Coefficient Stability: Theory and Evidence', Journal of Business & Economic Statistics **37**(2), 187–204. 20

Ravallion, M. (2020), Should the Randomistas (Continue to) Rule?, Working Paper 27554, National Bureau of Economic Research. 6

Romer, D. (2020), 'In Praise of Confidence Intervals', AEA Papers and Proceedings **110**, 55–60. 24

Rosenbaum, P. R. (2002), Observational Studies, Springer Series in Statistics, Springer New York, New York, NY. 20

Rosenthal, R. (1979), 'The file drawer problem and tolerance for null results', Psychological Bulletin **86**(3), 638–641. 2, 25

Rubin, D. B. (2001), 'Using Propensity Scores to Help Design Observational Studies: Application to the Tobacco Litigation', Health Services and Outcomes Research Methodology **2**(3), 169–188. 14

Schell, T. L., Griffin, B. A. and Morral, A. R. (2018), Evaluating Methods to Estimate the Effect of State Laws on Firearm Deaths: A Simulation Study, Technical report, RAND Corporation. 6

Shadish, W. R., Cook, T. D. and Campbell, D. T. (2002), Experimental and Quasi-experimental Designs for Generalized Causal Inference, Houghton Mifflin. 21, 24

Simmons, J. P., Nelson, L. D. and Simonsohn, U. (2011), 'False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant', Psychological Science **22**(11), 1359–1366. 24

Stommes, D., Aronow, P. M. and Sävje, F. (2021), 'On the reliability of published findings using the regression discontinuity design in political science', arXiv:2109.14526 [stat] . 5, 6, 21

Thistlethwaite, D. L. and Campbell, D. T. (1960), 'Regression-discontinuity analysis: An alternative to the ex post facto experiment', Journal of Educational Psychology **51**(6), 309–317. 7

Timm, A., Gelman, A. and Carlin, J. (2019), 'Retrodesign: Tools for Type S (Sign) and Type M (Magnitude) Errors'. 21

Young, A. (2021), 'Leverage, Heteroskedasticity and Instrumental Variables in Practical Application', p. 43. 6

Zwet, E. and Gelman, A. (2021), 'A Proposal for Informative Default Priors Scaled by the Standard Error of Estimates', The American Statistician pp. 1–9. 22

Zwet, E., Schwab, S. and Senn, S. (2021), 'The statistical properties of RCTs and a proposal for shrinkage', Statistics in Medicine **40**(27), 6107–6117. 22

Zwet, E. W. and Cator, E. A. (2021), 'The significance filter, the winner's curse and the need to shrink', Statistica Neerlandica **75**(4), 437–452. 2, 14, 22, 25, 31

A Mathematical proofs

I. 1 Variation of the exaggeration ratio (lemma 1)

Proof. Lu et al. (2019) and Zwet and Cator (2021) showed this in the case of $b = 0$. To extend it to the biased case, consider $E_b = \frac{\mathbb{E}\left[|\hat{\beta}_b| \ \middle| \beta_1, \sigma, |\hat{\beta}_b| > z_\alpha \sigma\right]}{|\beta_1|}$ the exaggeration ratio of interest. Note that, since $\hat{\beta}_b$ is an unbiased estimator of $\beta_1 + b$, $\tilde{E}_b = \frac{\mathbb{E}\left[|\hat{\beta}_b| \ \middle| \beta_1, \sigma, |\hat{\beta}_b| > z_\alpha \sigma\right]}{|\beta_{1+b}|}$ has the properties described in the lemma. Now, considering that $E_b = \left|\frac{\beta_1 + b}{\beta_1}\right| \tilde{E}_b$ proves the properties when β_1 and b have the same sign. □

I. 2 Asymptotic distribution of $\hat{\beta}_{\text{OVB}}$ (lemma 2)

Proof. Since, we do not observe w, we consider the projection of y on X only:

$$y = X\boldsymbol{\beta}_{\text{OVB}} + u_{\text{OVB}} \tag{5}$$

where by definition of the projection, $\mathbb{E}[X' u_{\text{OVB}}] = 0$.

We first compute the bias of the estimator. From equation 5 we get:

$$\begin{aligned}
& X'y = X'X\boldsymbol{\beta}_{\text{OVB}} + X'u_{\text{OVB}} \\
\Rightarrow \quad & \mathbb{E}[X'y] = \underbrace{\mathbb{E}[X'X]}_{\text{pos. def.}}\boldsymbol{\beta}_{\text{OVB}} + \underbrace{\mathbb{E}[X'u_{\text{OVB}}]}_{0} \\
\Leftrightarrow \quad & \boldsymbol{\beta}_{\text{OVB}} = \mathbb{E}[X'X]^{-1}\mathbb{E}[X'(X\boldsymbol{\beta} + \delta w + u)] \quad \text{cf eq. 2} \\
\Leftrightarrow \quad & \boldsymbol{\beta}_{\text{OVB}} = \boldsymbol{\beta} + \mathbb{E}[X'X]^{-1}\mathbb{E}[X'w]\delta
\end{aligned} \tag{6}$$

We then compute the asymptotic distribution. We can write:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) = \left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i \mathrm{x}_i'\right)^{-1} \sqrt{n}\left(\frac{1}{n}\sum_{i=1}^{n} \mathrm{x}_i u_{\text{OVB},i}\right)$$

Applying the Weak Law of Large Numbers (WLLN), the Central Limit Theorem (CLT) and Slutsky's theorem yields:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{OVB}} - \boldsymbol{\beta}_{\text{OVB}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[\mathrm{x}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i \mathrm{x}_i' u_{\text{OVB},i}^2]\mathbb{E}[\mathrm{x}_i \mathrm{x}_i']^{-1}\right) \tag{7}$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{x}_i \mathrm{x}_i']^{-1}$,

$\mathbb{E}[\mathrm{x}_i w_i]$ and $\mathbb{E}[\mathrm{x}_i \mathrm{x}_i' u_{\mathrm{OVB},i}^2]$.

$$\mathbb{E}[\mathrm{x}_i \mathrm{x}_i'] = \mathbb{E}\begin{bmatrix} 1 & x_i \\ x_i & x_i^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x \\ \mu_x & \sigma_x^2 + \mu_x^2 \end{bmatrix} \quad \Rightarrow \quad \mathbb{E}[\mathrm{x}_i \mathrm{x}_i']^{-1} = \frac{1}{\sigma_x^2}\begin{bmatrix} \sigma_x^2 + \mu_x^2 & -\mu_x \\ -\mu_x & 1 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{x}_i w_i] = \mathbb{E}\begin{bmatrix} w_i \\ x_i w_i \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbb{E}[x_i]\underbrace{\mathbb{E}[w_i]}_{0} + \mathrm{cov}(x_i, w_i) \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma \underbrace{\mathrm{var}(w_i)}_{\sigma_w^2} + \underbrace{\mathrm{cov}(\epsilon_i, w_i)}_{0} \end{bmatrix} = \begin{bmatrix} 0 \\ \gamma\sigma_w^2 \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{x}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i] = \frac{\gamma\sigma_w^2}{\sigma_x^2}\begin{bmatrix} -\mu_x \\ 1 \end{bmatrix} \tag{8}$$

Note that $\mathbb{E}[\mathrm{x}_i \mathrm{x}_i' u_{\mathrm{OVB},i}^2] \overset{\text{LIE}}{=} \mathbb{E}[\mathrm{x}_i \mathrm{x}_i' \mathbb{E}[u_{\mathrm{OVB},i}^2 | \mathrm{x}_i]]$. We thus first compute $\mathbb{E}[u_{\mathrm{OVB},i}^2 | \mathrm{x}_i]$, noting that:

$$\begin{aligned}
u_{\mathrm{OVB},i} &= y_i - \mathrm{x}_i' \boldsymbol{\beta}_{\mathrm{OVB}} \\
&= \delta w_i + u_i + \mathrm{x}_i'(\boldsymbol{\beta} - \boldsymbol{\beta}_{\mathrm{OVB}}) \\
&= \delta w_i + u_i - \underbrace{\mathrm{x}_i' \mathbb{E}[\mathrm{x}_i \mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{x}_i w_i]}_{\text{projection of } w_i \text{ on } x_i}\delta \\
&= u_i + \delta \underbrace{\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x) \right)}_{\text{part of } w_i \text{ orthogonal to } x_i} \\
&= u_i + \delta w_i^{\perp} \qquad\qquad \text{where } w_i^{\perp} = w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)
\end{aligned}$$

And thus,

$$\begin{aligned}
\mathbb{E}[u_{\mathrm{OVB},i}^2 | \mathrm{x}_i] &= \mathbb{E}[(u_i + \delta w_i^{\perp})^2 | x_i] \\
&= \mathbb{E}[u_i^2 | x_i] + 2\delta\mathbb{E}[u_i w_i^{\perp} | x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2 | x_i] \\
&= \sigma_u^2 + 2\delta\left(\mathbb{E}[u_i w_i | x_i] - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\underbrace{\mathbb{E}[u_i | x_i]}_{0} \right) + \delta^2\mathbb{E}[(w_i^{\perp})^2 | x_i] \\
&\overset{\text{LIE}}{=} \sigma_u^2 + 2\delta\mathbb{E}[w_i \underbrace{\mathbb{E}[u_i | x_i, w_i]}_{0} | x_i] + \delta^2\mathbb{E}[(w_i^{\perp})^2 | x_i] \\
&= \sigma_u^2 + \delta^2\mathbb{E}[(w_i^{\perp})^2 | x_i]
\end{aligned}$$

Notice that, by the law of total variance, $\mathbb{E}[(w_i^{\perp})^2 | x_i] = \mathrm{Var}(w_i^{\perp} | x_i) + \mathbb{E}[w_i^{\perp} | x_i]^2$. Now, since $w_i^{\perp}$ is the component of w_i that is orthogonal to x_i and by the projection interpretation of the

conditional variance, $\mathbb{E}[w_i^{\perp}|x_i] = 0$. And thus, since by assumption $\text{Var}(w_i^{\perp}|x_i) = \text{Var}(w_i^{\perp})$,

$$\begin{aligned}
\mathbb{E}[(w_i^{\perp})^2|x_i] &= \text{Var}(w_i^{\perp}|x_i) \\
&= \text{Var}(w_i^{\perp}) \\
&= \mathbb{E}[(w_i^{\perp})^2] - \mathbb{E}[w_i^{\perp}]^2 \\
&= \mathbb{E}\left[\left(w_i - \frac{\gamma\sigma_w^2}{\sigma_x^2}(x_i - \mu_x)\right)^2\right] - \left(\underbrace{\mathbb{E}[w_i]}_{0} + \frac{\gamma\sigma_w^2}{\sigma_x^2}\underbrace{\mathbb{E}[x_i - \mu_x]}_{0}\right)^2 \\
&= \underbrace{\mathbb{E}[w_i^2]}_{\sigma_w^2} - 2\frac{\gamma\sigma_w^2}{\sigma_x^2}\left(\underbrace{\mathbb{E}[x_i w_i]}_{\gamma\sigma_w^2} - \mu_x \underbrace{\mathbb{E}[w_i]}_{0}\right) + \frac{\gamma^2\sigma_w^4}{\sigma_x^4}\underbrace{\mathbb{E}[(x_i - \mu_x)^2]}_{\sigma_x^2} \\
&= \sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)
\end{aligned}$$

Note that this variance is well defined (positive) only if $\sigma_x^2 \geq \gamma^2\sigma_w^2$. Under this condition,

$$\mathbb{E}[u_{\text{OVB},i}^2|\text{x}_i] = \sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right) \tag{9}$$

Thus, under our set of assumptions, $\mathbb{E}[u_{\text{OVB},i}^2|\text{x}_i]$ does not depend on x_i and $\mathbb{E}[u_{\text{OVB},i}^2|\text{x}_i] = \mathbb{E}[u_{\text{OVB},i}^2]$. We denote this quantity $\sigma_{u_{\text{OVB}}}^2$.

We can now compute the variance of the estimator $\hat{\boldsymbol{\beta}}_{\text{OVB}}$, noting that $\mathbb{E}[\text{x}_i\text{x}_i' u_{\text{OVB},i}^2] = \mathbb{E}[\text{x}_i\text{x}_i'\mathbb{E}[u_{\text{OVB},i}^2|\text{x}_i]] = \mathbb{E}[\text{x}_i\text{x}_i'\sigma_{u_{\text{OVB}}}^2] = \sigma_{u_{\text{OVB}}}^2\mathbb{E}[\text{x}_i\text{x}_i']$. And thus $\mathbb{E}[\text{x}_i\text{x}_i']^{-1}\mathbb{E}[\text{x}_i\text{x}_i' u_{\text{OVB},i}^2]\mathbb{E}[\text{x}_i\text{x}_i']^{-1} = \sigma_{u_{\text{OVB}}}^2\mathbb{E}[\text{x}_i\text{x}_i']$.

Plugin this and equation 8 into equation 7, we get, for $\hat{\beta}_{\text{OVB}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{OVB}}$:

$$\hat{\beta}_{\text{OVB}} \xrightarrow{d} \mathcal{N}\left(\beta_1 + \frac{\delta\gamma\sigma_w^2}{\sigma_x^2},\ \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \frac{\gamma^2\sigma_w^2}{\sigma_x^2}\right)}{n\ \sigma_x^2}\right)$$

Then, noting that $\rho_{xw} = \text{corr}(x, w) = \frac{\text{cov}(\mu_x + \gamma w + \epsilon, w)}{\sigma_x\sigma_w} = \frac{\gamma\sigma_w}{\sigma_x}$, we have:

$$\sigma_{\text{OVB}}^2 = \text{avar}\left(\hat{\beta}_{\text{OVB}}\right) = \frac{\sigma_u^2 + \delta^2\sigma_w^2\left(1 - \rho_{xw}^2\right)}{n\ \sigma_x^2}$$

□

I. 3 Asymptotic distribution of $\hat{\beta}_{\text{CTRL}}$ (lemma 3)

Proof. The proof is the well know proof of the asymptotic distribution of the OLS. I simply compute $\mathbb{E}[x_{w,i}x'_{w,i}]^{-1}$ to retrieve the variance of the parameter of interest $\boldsymbol{\beta}_{\text{CTRL}}$. We know that we have:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{CTRL}} - \boldsymbol{\beta}_{\text{CTRL}}) \xrightarrow{d} \mathcal{N}\left(0, \mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]^{-1}\sigma_u^2\right)$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{CTRL}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]^{-1}$.

$$\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}] = \mathbb{E}\begin{bmatrix} 1 & x_i & w \\ x_i & x_i^2 & x_i w_i \\ w_i & x_i w_i & w_i^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_x & 0 \\ \mu_x & \sigma_x^2 + \mu_x^2 & \gamma\sigma_w^2 \\ 0 & \gamma\sigma_w^2 & \sigma_w^2 \end{bmatrix}$$

Note that we have $\mathbb{E}[x_i w_i] = \mathbb{E}[x_i]\underbrace{\mathbb{E}[w_i]}_{0} + \text{cov}(x_i, w_i) = \gamma\underbrace{\text{var}(w_i)}_{\sigma_w^2} + \underbrace{\text{cov}(\epsilon_i, w_i)}_{0} = \gamma\sigma_w^2$.

Now, $\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]^{-1} = \dfrac{1}{\det(\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}])}\,{}^t\mathrm{C}$ with C the comatrix of $\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]$. We have:

$$\det(\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]) = (\sigma_x^2 + \mu_x^2)\sigma_w^2 - \sigma_w^2\mu_x^2 - \gamma^2\sigma_w^4 = \sigma_w^2(\sigma_x^2 - \gamma^2\sigma_w^2)$$

and the "central" component of C, σ_w^2. Thus the central component of interest of $\mathbb{E}[\mathrm{x}_{w,i}\mathrm{x}'_{w,i}]^{-1}$ is $\frac{1}{\sigma_x^2 - \gamma^2\sigma_w^2}$. Therefore, for $\hat{\beta}_{\text{CTRL}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{CTRL}}$, we have:

$$\hat{\beta}_{\text{CTRL}} \xrightarrow{d} \mathcal{N}\left(\beta_1,\ \frac{\sigma_u^2}{n\ (\sigma_x^2 - \gamma^2\sigma_w^2)}\right) \tag{10}$$

Then, noting that $\rho_{xw} = \text{corr}(x, w) = \frac{\text{cov}(\mu_x + \gamma w + \epsilon, w)}{\sigma_x\sigma_w} = \frac{\gamma\sigma_w}{\sigma_x}$, we have:

$$\sigma^2_{\text{CTRL}} = \frac{\sigma_u^2}{n\ \sigma_x^2(1 - \rho_{xw}^2)}$$

□

I. 4 Asymptotic distribution of $\hat{\beta}_{\text{IV}}$ (lemma 4)

Proof. Since $u_{\text{IV}} = u_{\text{OVB}} = \delta w + u$, we have $\sigma^2_{u_{\text{IV}}} = \sigma_u^2 + \delta^2\sigma_w^2$. Thus, the usual asymptotic distribution of the IV gives:

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{IV}} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}\left(0, (\sigma_u^2 + \delta^2\sigma_w^2)\mathbb{E}[\mathrm{z}_i\mathrm{x}'_i]^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}'_i]\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}'_i]^{-1}\right)'\right)$$

We are interested in the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$. To retrieve it we need to compute $\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']$, $\mathbb{E}[\mathrm{x}_i\mathrm{z}_i']^{-1}$ and its transpose.

$$\mathbb{E}[\mathrm{z}_i\mathrm{z}_i'] = \mathbb{E}\begin{bmatrix} 1 & z_i \\ z_i & z_i^2 \end{bmatrix} = \begin{bmatrix} 1 & \mu_z \\ \mu_z & \sigma_z^2 + \mu_z^2 \end{bmatrix}$$

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i'] = \begin{bmatrix} 1 & \mathbb{E}[x_i] \\ \mathbb{E}[z_i] & \mathbb{E}[z_i x_i] \end{bmatrix} = \begin{bmatrix} 1 & \pi_0 + \pi_1\mathbb{E}[z_i] + \gamma\underbrace{\mathbb{E}[w_i]}_{0} + \underbrace{\mathbb{E}[e_i]}_{0} \\ \mu_z & \pi_0\mathbb{E}[z_i] + \pi_1\mathbb{E}[z_i^2] + \gamma\underbrace{\mathbb{E}[z_i w_i]}_{0} + \underbrace{\mathbb{E}[z_i e_i]}_{0} \end{bmatrix} = \begin{bmatrix} 1 & \pi_0 + \pi_1\mu_z \\ \mu_z & \pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) \end{bmatrix}$$

$$\Rightarrow \quad \mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1} = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} \pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) & -\pi_0 - \pi_1\mu_z \\ -\mu_z & 1 \end{bmatrix}$$

Thus,

$$\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\mathbb{E}[\mathrm{z}_i\mathrm{z}_i']\left(\mathbb{E}[\mathrm{z}_i\mathrm{x}_i']^{-1}\right)' = \frac{1}{\pi_1\sigma_z^2}\begin{bmatrix} 2\pi_0\mu_z + \pi_1(\sigma_z^2 + \mu_z^2) + \frac{\pi_0^2}{\pi_1} & -\mu_z - \frac{\pi_0}{\pi_1} \\ -\mu_z - \frac{\pi_0}{\pi_1} & \frac{1}{\pi_1} \end{bmatrix}$$

And so, for $\hat{\beta}_{\text{IV}}$, the second component of $\hat{\boldsymbol{\beta}}_{\text{IV}}$, we have:

$$\sqrt{n}\left(\hat{\beta}_{\text{IV}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma_u^2 + \delta^2\sigma_w^2}{n\ \pi_1^2\sigma_z^2}\right) \tag{11}$$

Now, since $\rho_{xz} = \text{corr}(x_i, z_i) = \frac{\text{cov}(\pi_0 + \pi_1 z_i + \gamma w_i + e_i, z_i)}{\sigma_x\sigma_z} = \pi_1\frac{\sigma_z}{\sigma_x}$,

$$\sqrt{n}\left(\hat{\beta}_{\text{IV}} - \beta_1\right) \xrightarrow{d} \mathcal{N}\left(0,\ \frac{\sigma_u^2 + \delta^2\sigma_w^2}{\sigma_x^2\rho_{xz}^2}\right)$$

□

I. 5 Asymptotic distribution of $\hat{\beta}_{\text{RED}}$ (lemma 5)

Proof. The proof is straightforward: this is the usual univariate, unbiased case, with and error term equal to $(\delta + \beta_1\gamma)w_i + u_i + \beta_1\epsilon_{\text{RED},i}$. Since w, u and ϵ_{RED} uncorrelated, its variance is $(\delta + \beta_1\gamma)^2\sigma_w^2 + \sigma_u^2 + \beta_1^2\sigma_{\epsilon_{\text{RED}}}^2$. □